Consistency
Discord ✓
Help ✓
Twitter ✗ ✓
#CoderArmy #dp20

Segment ✓
Trie ✓

Dynamic Programming

→ Jo kaam ek baar liya
→ usko dobara mat karo

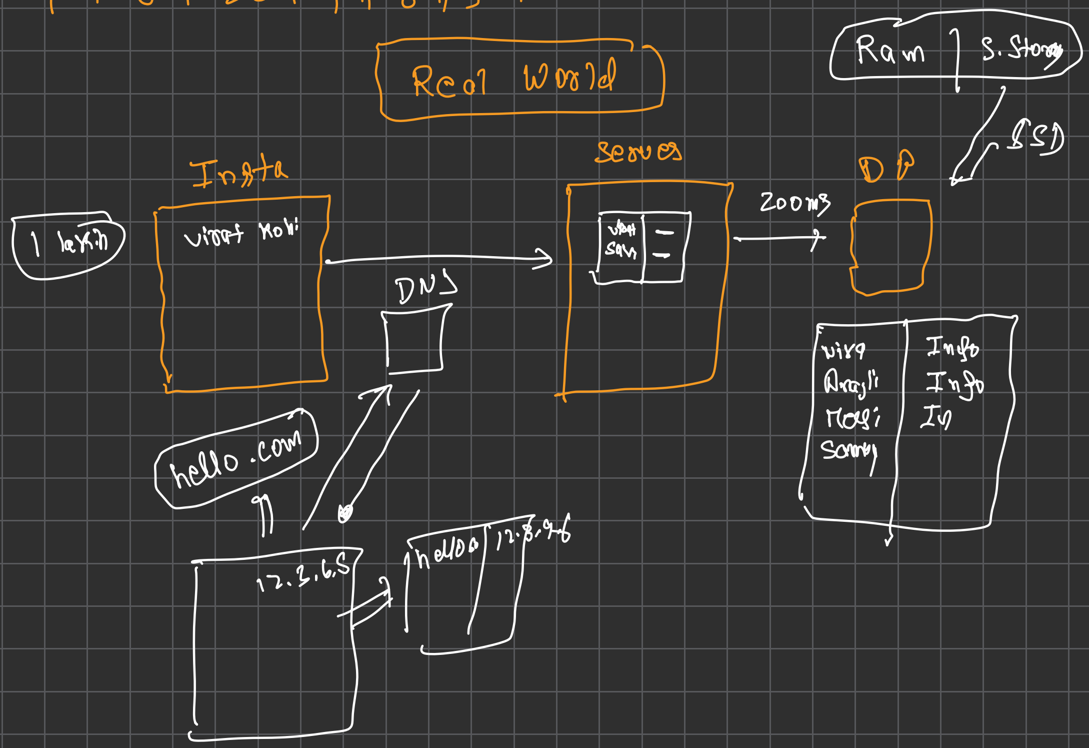
6 + 20 + 9 ⇒ 35

6 + 20 + 9 + 8 + 5 ⇒ 48

7 + 6 + 20 + 9 + 8 + 5 + 2 ⇒ 57

Energy
↳ CPU
Electricity

Real World



Chatgpt / claude

what is the capital of India

Capital of India is new Delhi

$20 140 93 82

701 809

what is the capital of India

56 70 18 23 90 48

KV Cache

2 3 5

Query
Key
Value

50 70 18 23 90 48 120 140

Q / K \ V

2 3 5

100 150 250

140 | 210 | 350

96 | 144 | 240

240 | 360 | 600

[- - . - -] × [ ]

4096

4096×4096

4 6 8 10 12 14 16 18 20 222

1100 1200 1300 1400 1500 1600 1700 1800

Made with Goodnotes

1 2 4 7 11 16

$\hookrightarrow$ 7 8 16 32

$$[2, 3, 4, 9] \times \begin{bmatrix} 4 & 3 & 6 & 5 \\ 2 & 8 & 9 & 3 \\ 1 & 2 & 1 & 9 \\ 6 & 4 & 8 & 2 \end{bmatrix}$$

1, 1, 2

[Computation → calculation ⇒ fast
ek baar kar liya, don't repeat]

[Fibonnaci Series]

0	1	2	3	4	5	6	7	8	
0	1	1	2	3	5	8	13	21	. . . . . .

5   5%   10%

Recursive → Top down → Bottom up → Space optimal

80%

n = 5   6

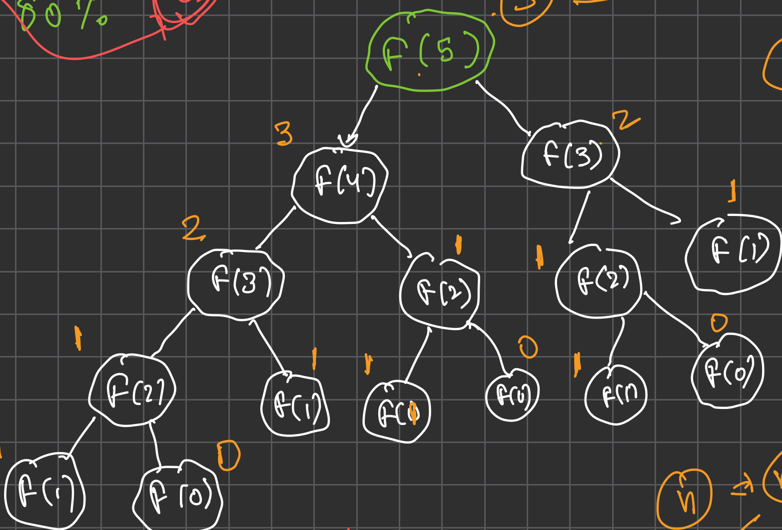
f(5)
f(4) ⇒ 3
f(3) = 2
f(2) = 1
f(1) ⇒ 1
f(0) ⇒ 0

answer ⇒ 5

0 → 0
1 → 1

n ⇒ 5

2^n



n ⇒ (n+1)

0	1	2	3	4	5
0	1	1	2	3	5

dp

⇒ Bottom-up
↳ LOOP

① Base Cases
②

0 − 0
1 − 1

dp[i] = dp[i-1] + dp[i-2]

result
answer

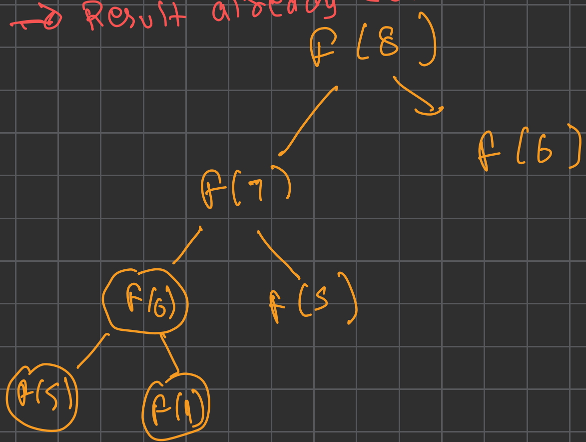
```
int fib(int n, vector<int>&dp)
{
    if(n<=1)
        return n
    if(dp[n]!=-1)
        return dp[n];
    return dp[n] = fib(n-1) + fib(n-2)
    dp[i] = dp[i-1] + dp[i-2]
}
```

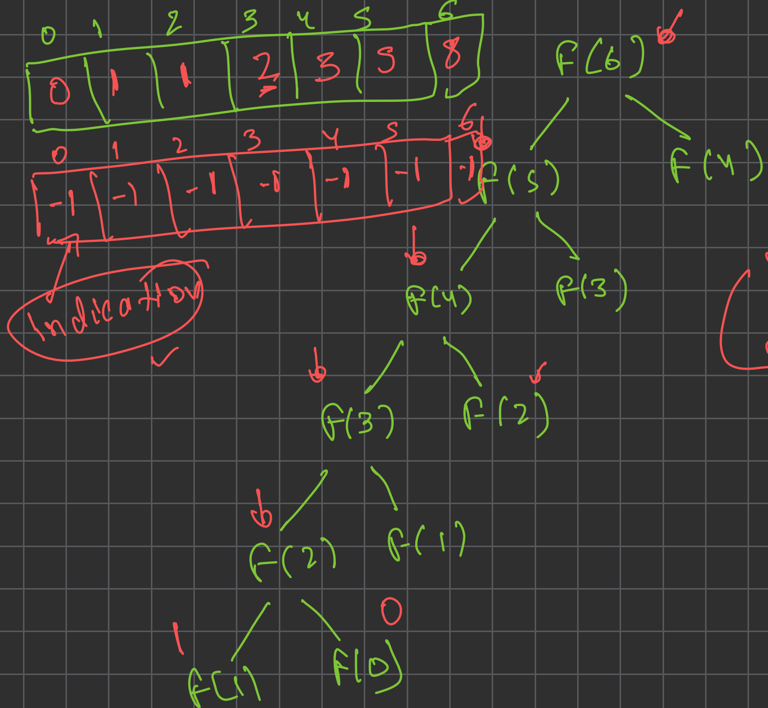
Top down approach

Recursive code

↳ Result store

→ Result already calculated hai, dubara calculate mat karo





7

$\rightarrow O(n)$

2^n

$O(n)$

$O(n)$

Space comp $O(n)$

for loop se solve

↳ vector $O(n)$

Time Comp $O(n)$

F [1 tak]

memoization

0	1	2	3	4	5
0	1	1	2	3	5

0	1	2
1	1	2

$O(n)$

$O(1)$

```
for ( i = 3 ; i <= n ; i++ )
{
    dp[0] = dp[1]
    dp[1] = dp[2]
    dp[2] = dp[0] + dp[1]
}
```